Scope: what a scaling ladder predicts

> **Reading note.** This is an explanatory reconstruction of Jiaxuan Zou's literature synthesis. Published experiments belong to the cited teams. Numbered protocols and illustrative matrices are our planning synthesis unless explicitly attributed.

A scaling ladder is a designed collection of training runs. Each run varies a known quantity, while the remaining recipe is fixed or changes according to a declared rule. The blog's §1 distinguishes dense, mixture-of-experts (MoE), and data-constrained settings. This matters because a formula fitted to one setting need not apply to another.

Questions and outputs

The central objects are model size N, training tokens D, compute C, and held-out loss L. A ladder can predict loss, select N versus D under a compute budget, compare recipes, or transfer hyperparameters. These are different estimation problems. A prescribed recipe can be evaluated without independently finding the best hyperparameters at every point; claiming the optimal frontier requires the latter effort.

For a dense model, distinguish trunk parameters N_{body} from embedding and head parameters. For MoE, record total and per-token active counts. For finite data, distinguish loss-bearing tokens D from unique tokens U. These definitions determine what a fitted exponent means.

Procedure

1. Write one target decision: for example, select model size under a fixed training budget.
2. Specify dense or MoE, intended token-to-parameter ratio, available unique data, context length, and deployment constraints.
3. Name the predicted quantity and its units. Cross-entropy in nats/token is different from bits/byte and benchmark accuracy.
4. Specify an extrapolation target and the largest fitting point. Reserve runs beyond that point for validation.

A useful starting relation is $C \approx 6ND$, where C is training FLOPs and N is a compatible dense parameter count. It is an approximation; the measurement guide explains the omitted attention and head costs.

Failure to watch

A smooth curve through small models establishes a fit, not a reliable large-model forecast. Delphi's failed initial recipe is the blog's motivating example; its stability failure is evidence for testing beyond the fitting regime, not evidence for a universal maximum extrapolation factor.

Deliverable: a one-page scope statement with the target, model/data regime, measurement definitions, and intended decision.

Sources: blog §1; Marin, Delphi report.

Blog/notes crosswalk · Detailed companion